Year 13 Maths - Differential equations

Similar questions - Cooling, draining tanks and separation of variables

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A cup of tea is made at 90°C in a room at a constant temperature of 15°C. After 5 minutes its temperature is 60°C. Let θ°C be its temperature after t minutes.

(a) In a simple model the temperature falls at a constant rate. Find θ in terms of t and the time at which this model predicts $\theta = 30$. **[3]**

(b) In a better model the rate of decrease of θ is proportional to $\theta - 15$. Write down a differential equation and show that $\theta = 15 + 75\mathrm{e}^{-kt}$ where $k = \frac{1}{5}\ln\frac{5}{3}$. **[6]**

(c) Find the difference, to the nearest minute, between the two predictions of when the tea reaches 30°C. **[3]**

2 A cylindrical tank is initially full to a depth of 1 m. Water leaks from a hole in the base and the depth x m after t seconds is modelled by $\dfrac{\mathrm{d}x}{\mathrm{d}t} = -k\sqrt{x}$ where k is a constant. When $x = 0.64$ the depth is decreasing at $0.004\,\mathrm{m\,s^{-1}}$.

(a) Show that $k = 0.005$. **[2]**

(b) Solve the differential equation to find x in terms of t. **[4]**

(c) Find the time at which the tank empties. **[2]**

3 The gradient of a curve is given by $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2x\ln x}{\mathrm{e}^{2y}}$ and the curve passes through $(1,0)$.

(a) Find the equation of the curve in the form $\mathrm{e}^{2y} = \mathrm{f}(x)$. **[6]**

(b) Find the exact value of y when $x = \mathrm{e}$. **[2]**

Question 1

(a) Rate $= \frac{90-60}{5} = 6°$C per minute, $\theta = 90 - 6t$. $\theta = 30$ at $t = 10$ minutes

(b) $\dfrac{\mathrm{d}\theta}{\mathrm{d}t} = -k(\theta - 15)$. $\displaystyle\int \frac{\mathrm{d}\theta}{\theta - 15} = -\int k\,\mathrm{d}t$ gives $\ln(\theta - 15) = -kt + c$, so $\theta = 15 + A\mathrm{e}^{-kt}$. $t = 0$: $A = 75$. $t = 5$: $60 = 15 + 75\mathrm{e}^{-5k}$, $\mathrm{e}^{-5k} = \frac{3}{5}$, so $k = \frac{1}{5}\ln\frac{5}{3}$

(c) $30 = 15 + 75\mathrm{e}^{-kt}$ gives $\mathrm{e}^{-kt} = \frac{1}{5}$, $t = \dfrac{\ln 5}{k} = 15.7$ minutes. Difference $= 15.7 - 10 \approx 6$ minutes

Question 2

(a) $-0.004 = -k\sqrt{0.64} = -0.8k$, so $k = 0.005$

(b) $\int x^{-1/2}\mathrm{d}x = -0.005\int \mathrm{d}t$ gives $2\sqrt{x} = -0.005t + c$. $t = 0$, $x = 1$: $c = 2$. $\sqrt{x} = 1 - 0.0025t$, $x = (1 - 0.0025t)^2$

(c) $x = 0$ when $t = 400$ seconds

Question 3

(a) $\int \mathrm{e}^{2y}\mathrm{d}y = \int 2x\ln x\,\mathrm{d}x$. Left: $\frac{1}{2}\mathrm{e}^{2y}$. Right (by parts): $x^2\ln x - \frac{1}{2}x^2 + c$. At $(1,0)$: $\frac{1}{2} = -\frac{1}{2} + c$, $c = 1$. $\mathrm{e}^{2y} = 2x^2\ln x - x^2 + 2$

(b) $\mathrm{e}^{2y} = 2\mathrm{e}^2 - \mathrm{e}^2 + 2 = \mathrm{e}^2 + 2$, so $y = \frac{1}{2}\ln(\mathrm{e}^2 + 2)$